Gabarito Lista 4 - MATF14
Estatística Econômica I

2025.2

Questão 1

Derivadas das funções:

- $f(x) = 5 \Rightarrow f'(x) = 0$
- $g(x) = x^6 \Rightarrow g'(x) = 6x^5$
- $h(x) = x^{15} \Rightarrow h'(x) = 15x^{14}$

Questão 2

Derivadas das funções:

(a) $f(x) = 8x^{11} \Rightarrow f'(x) = 88x^{10}$

(b) $f(x) = -\frac{7}{5}x^3 - \frac{\sqrt{3}}{7} \Rightarrow f'(x) = -\frac{7}{5} \cdot 3x^2 = -\frac{21}{5}x^2$

(c) $f(x) = 5 + x + 3x^2 \Rightarrow f'(x) = 0 + 1 + 6x = 1 + 6x$

(d) $f(x) = 3+2x^n+x^{2n} \Rightarrow f'(x) = 0+2nx^{n-1}+2nx^{2n-1} = 2nx^{n-1}+2nx^{2n-1}$

(e) $f(x) = e^{-3x} \Rightarrow f'(x) = e^{-3x} \cdot (-3) = -3e^{-3x}$

(f) $f(x) = e^{7x^2-2x} \Rightarrow f'(x) = e^{7x^2-2x} \cdot (14x - 2) = (14x - 2)e^{7x^2-2x}$

Questão 3

Primitivas das funções:

(a) $f(x) = \sqrt{x} = x^{1/2} \Rightarrow F(x) = \frac{x^{3/2}}{3/2} = \frac{2}{3}x^{3/2} + C$

(b) $f(x) = \frac{1}{x^3} = x^{-3} \Rightarrow F(x) = \frac{x^{-2}}{-2} = -\frac{1}{2x^2} + C$

(c) $f(x) = x^{-2/5} \Rightarrow F(x) = \frac{x^{3/5}}{3/5} = \frac{5}{3}x^{3/5} + C$

Questão 4

Integrais definidas:

(a) $\displaystyle\int_0^1 7\,dx = [7x]_0^1 = 7(1) - 7(0) = 7$

(b) $\displaystyle\int_1^2 x^3\,dx = \left[\frac{x^4}{4}\right]_1^2 = \frac{16}{4} - \frac{1}{4} = \frac{15}{4} = 3,75$

(c) $\displaystyle\int_0^1 (-x^2)\,dx = \left[-\frac{x^3}{3}\right]_0^1 = -\frac{1}{3} - 0 = -\frac{1}{3}$

(d) $\displaystyle\int_0^2 (x^2 - 3x + 5)\,dx = \left[\frac{x^3}{3} - \frac{3x^2}{2} + 5x\right]_0^2$
$= \left(\frac{8}{3} - \frac{12}{2} + 10\right) - 0 = \frac{8}{3} - 6 + 10 = \frac{8}{3} + 4 = \frac{20}{3}$

(e) $\displaystyle\int_0^2 x(x^5 - 1)\,dx = \int_0^2 (x^6 - x)\,dx = \left[\frac{x^7}{7} - \frac{x^2}{2}\right]_0^2$
$= \left(\frac{128}{7} - \frac{4}{2}\right) - 0 = \frac{128}{7} - 2 = \frac{128-14}{7} = \frac{114}{7}$

(f) $\displaystyle\int_0^3 e^{2x}\,dx = \left[\frac{e^{2x}}{2}\right]_0^3 = \frac{e^6}{2} - \frac{e^0}{2} = \frac{e^6 - 1}{2}$